

EduSaaS PDDD - Part 3

Chapters 12 to 17: Enums, Constraints, FKs, Indexes, Triggers and Functions

Document Type: Production Database Design Document (PDDD)

Schema: education | Database: PostgreSQL | Architecture: Backend-owned persistence, stateless AI services

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Chapter 12: Enum Catalogue

Enum	Values	Used By
community_post_status	Draft, Published, Archived, Flagged	community_posts.status
community_post_type	Discussion, Question, Course, Announcement, Job, Internship, Project, Achievement, Event	community_posts.post_type
community_reaction_type	Like, Celebrate, Support, Insightful	community_reactions.reaction_type
community_visibility	Public, Student, Educators, Employers, Admins	community_posts.visibility

Chapter 13: Complete Constraint Catalogue

Primary keys exist on all 46 tables. Composite keys exist on role_permissions, quiz_state and community_post_tags. Unique constraints protect roles, permissions, users.email, profiles.user_id, enrollments(user_id, course_id), applications(job_id, student_id), certificates.certificate_code, skills.skill_name, domain_roles.domain_name, difficulty values, community tags, bookmarks and settings(scope, key).

Recommended Check	Purpose
completion_percentage between 0 and 100	Progress/enrollment correctness
rating between 1 and 5	Course ratings validation
course_id <> prerequisite_course_id	Prevent self prerequisite
confidence_score between 0 and 1	Recommendation quality
end_date > start_date	Subscription validity
exactly one of post_id/comment_id in reactions	Prevent ambiguous reactions

Chapter 14: Foreign Key Catalogue

Child Column	Parent Column
users.role_id	roles.id
users.domain_role_id	domain_roles.domain_role_id
profiles.user_id	users.id
courses.educator_id	users.id
lessons.course_id	courses.id
enrollments.user_id	users.id
enrollments.course_id	courses.id
progress.user_id	users.id
progress.lesson_id	lessons.id
recommendations.user_id	users.id
recommendations.course_id	courses.id
questions.skill_id	skills.skill_id
questions.difficulty_id	difficulty_levels.difficulty_id
quiz_sessions.user_id	users.id
student_answers.session_id	quiz_sessions.session_id
jobs.employer_id	users.id
applications.job_id	jobs.id
applications.student_id	users.id
community_posts.author_id	users.id
community_comments.post_id	community_posts.id
community_reactions.user_id	users.id
notifications.user_id	users.id
activity_logs.user_id	users.id
course_skills.course_id	courses.id
course_skills.skill_id	skills.skill_id
course_ratings.user_id	users.id

Chapter 15: Index Catalogue

Domain	Key Indexes
Security	users_email_key, idx_users_role, users_domain_role_id_idx
Learning	idx_courses_educator, idx_enrollment_user, idx_progress_user, idx_recommendations_user
Assessment	idx_assessment_user, idx_quiz_sessions_user, idx_student_answers_session, idx_student_skill_results_skill
Employer	idx_jobs_employer, idx_application_job, idx_application_student
Community	idx_community_posts_author, idx_community_posts_created_at, idx_reaction_user, idx_bookmark_user
Administration	idx_notification_user, idx_subscription_user, idx_announcements_educator
AI Support	idx_activity_logs_user_last, idx_course_skills_skill, idx_course_ratings_course

Recommended Composite Indexes

```
CREATE INDEX idx_recommendation_user_rank ON education.recommendations(user_id, rank);
CREATE INDEX idx_application_job_status ON education.applications(job_id, status);
CREATE INDEX idx_questions_skill_difficulty ON education.questions(skill_id, difficulty_id);
CREATE INDEX idx_notification_unread ON education.notifications(user_id, read_status);
CREATE INDEX idx_activity_logs_user_last ON education.activity_logs(user_id, last_activity);
```

Chapter 16: Trigger Catalogue

```
CREATE OR REPLACE FUNCTION education.set_updated_at() RETURNS TRIGGER AS $$
BEGIN
    NEW.updated_at = CURRENT_TIMESTAMP;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;
```

Trigger	Table	Type
trg_progress_updated_at	progress	BEFORE UPDATE
trg_gap_reports_updated_at	gap_reports	BEFORE UPDATE
trg_community_comments_updated_at	community_comments	BEFORE UPDATE
trg_activity_logs_updated_at	activity_logs	BEFORE UPDATE
trg_login_history_updated_at	login_history	BEFORE UPDATE
trg_course_skills_updated_at	course_skills	BEFORE UPDATE
trg_course_prerequisites_updated_at	course_prerequisites	BEFORE UPDATE
trg_course_ratings_updated_at	course_ratings	BEFORE UPDATE
trg_recommendations_updated_at	recommendations	BEFORE UPDATE

Chapter 17: Functions Catalogue

Function	Purpose
education.set_updated_at()	Maintain updated_at audit field
fn_increment_comment_count()	Maintain post comment counts
fn_decrement_comment_count()	Maintain post comment counts
fn_course_completion_percentage()	Calculate course completion
fn_readiness_score()	Calculate readiness score
fn_validate_prerequisites()	Validate learning prerequisites
fn_generate_certificate_code()	Generate certificate code